HSC BOARD EXAM 2024 - PAST PAPER & SOLUTION

BIOLOGY SECOND PAPER (ZOOLOGY) - SOLVED QUESTIONS

1. Board Creative Question 1 (Circulatory System)

Stimulus: A 52-year-old individual experienced severe retrosternal chest pain radiating to the left arm. Angiography showed 85% occlusion of the left anterior descending coronary artery.

Questions:

a) What is angioplasty? (1 Mark)

b) Distinguish between systemic circulation and pulmonary circulation. (2 Marks)

c) Explain the cardiac cycle events corresponding to heart sound generation (Lub and Dub). (3 Marks)

d) Compare coronary stenting and coronary artery bypass graft (CABG) for relieving the condition. (4 Marks)

2. Board Creative Question 2 (Renal Excretion)

Stimulus: The renal cortex contains 1 to 1.2 million functional units responsible for selective reabsorption and urine formation.

Questions:

a) What is podocyte? (1 Mark)

b) Explain the role of antidiuretic hormone (ADH) in osmoregulation. (2 Marks)

c) Describe the mechanism of glomerular ultrafiltration based on effective filtration pressure. (3 Marks)

d) 'Kidney dialysis serves as an artificial life support' - Justify this statement with clinical principles. (4 Marks)

3. Board Multiple Choice Analysis (2024)

1. Which chamber of the human heart has the thickest muscular wall?
   (A) Right atrium (B) Left atrium (C) Right ventricle (D) Left ventricle
   Answer: (D) Left ventricle pumps systemic blood at high pressure.

2. Which immunoglobulin is predominantly present in maternal colostrum milk?
   (A) IgG (B) IgA (C) IgM (D) IgE
   Answer: (B) Secretory IgA provides mucosal neonatal immunity.

3. What is the phenotypic ratio of dominant epistasis in dihybrid crosses?
   (A) 9:3:3:1 (B) 12:3:1 (C) 9:7 (D) 13:3
   Answer: (B) 12:3:1.